



Dynamics of marine bacterial biofouling communities after initial *Alteromonas genovensis* biofilm attachment to anti-fouling paint substrates

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ABSTRACT

To determine how bacterial communities succeed after the initial attachment of the bacterial biofilm adhesion using 16S rDNA meta-barcoding in plates coated with copper-based anti-fouling (AF) and non-AF (control) coatings as well as ambient seawater, coated plates were submerged in a marine environment in situ. *Alteromonas genovensis* (Gammaproteobacteria) in AF coating and *Pacificibacter* sp. (Alphaproteobacteria) in the control plate were initially abundant. In the AF coating, the abundance of *A. genovensis* decreased rapidly, whereas that of genus *Phaeobacter* (Alphaproteobacteria), *Serratia* (Gammaproteobacteria) and *Cupriavidus* (Betaproteobacteria) increased. Bacterial community in the control plate had a strong connection to pathogenic *Vibrio* spp. associated with the growth of invertebrates. Therefore, in the in situ AF coating experiment, *A. genovensis* accumulation was initially and intensively increased, and the bacteria responded to chemical antagonism, induced the proliferation of specific biofilm bacteria and influenced the interactions and recruitment of additional bacterial communities.

1. Introduction

Biofouling is caused by living organisms, including bacteria, diatoms, fungi, protozoans, macroalgae and/or invertebrates, that are attached to submerged surfaces (Dobretsov et al., 2013). Bacterial communities, which are primary colonisers, initiate biofilm formation and induce the secondary attachment of diatoms (Cooksey and Wigglesworth-Cooksey, 1995). Specific macroalgae and invertebrates eventually start growing on the substrate (Mieszkin et al., 2013). Extracellular polymeric substances secreted by some marine heterotrophic bacteria lead to the formation of the biofilm and enable these bacteria to attach to natural or artificial substrates (Harder et al., 2002; Wieczorek et al., 1995). The stimulation or inhibition of biofouling bacteria is influenced by the unique characteristics of the substrate, including viscosity, surface tension and the ability to fulfil the energy requirements of the bacteria (Zhao et al., 2012). Moreover, some pathogenic bacteria were present in the biofilm of a ship's hull. Inbakandan et al. (2010) found that the opportunistic pathogen *Proteus mirabilis* was present in the anti-fouling (AF) coating of a ship's hull, and Shikuma and

Hadfield (2010) reported that the potential human pathogenic bacteria *Escherichia coli* and *Vibrio cholera* are found in harbours and in the hulls of ships and form microbial films on submerged surfaces.

Hull fouling is considered an economic and environmental hazard. Schultz et al. (2011) reported that damage caused by the movement of a ship due to the drag attributed to fouling organisms was estimated to be approximately US\$ 56 million per year. Another significant problem is ecological disturbances caused by the spread of invasive species transported via ship hulls (Mack et al., 2000; Simberloff, 2006). AF coatings are special coatings that prevent the accumulation of marine organisms on submerged surfaces. AF coating comprises biocides, which are used either alone or in combination with heavy metals. In particular, copper has been used as the major active biocide in AF coating (Yebra et al., 2004). When heavy metals accumulate above certain concentrations, they exert adverse effects on the reproduction and growth of the attached organisms on the hull (Winfield et al., 2018). However, this copper-based AF coating does not exhibit its properties permanently. Casse and Swain (2006) reported the loss of activity of the copper-based AF coating over time, i.e. 15–60 days.

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Meta-barcoding approaches enable the characterisation of microorganisms as well as the analysis of complex environmental samples. Several ecological studies on microorganisms using 16S rDNA meta-barcoding have estimated changes in population dynamics (Jung et al., 2021; Kang et al., 2021; Kim et al., 2016). Furthermore, some studies that characterised bacteria and attached eukaryotes used meta-barcoding or metagenomic approaches (Pochon et al., 2015; Zaiko et al., 2016). In this study, we assessed the dynamics of a bacterial community present on AF-coated artificial plates and compared them with plates coated with a non-AF (control) agent and with ambient seawater in situ using 16S rDNA meta-barcoding. In particular, we focused on the relationships between bacterial communities present on AF coating and its environment, specific responses by bacteria (or key bacterial taxa with specific adaptations) on the coating and potential pathogenic bacteria harboured on the coating.

2. Material and methods

2.1. In situ experimental design of AF coating

The in situ experiment involving AF hull coatings was performed at the Jangmok Bay Time-series Monitoring Site (JB-TMS, 34° 59' 37''N and 128° 40' 27''E) on the southern coast of South Korea for 49 days from 8 May to 26 June 2018 (Supplementary Fig. S1). The JB-TMS sampling site has been described previously by Kim et al. (2016). The iron plate that mimicked the ship's hull surface and measured 200 × 300 mm was painted twice on both sides using a roller. The experiments were performed in two groups. The first group comprised the non-AF (control) plate coated with commercially available paint (Tie-Coat Primer, Permanent Painted Coatings Ltd., USA). The second group comprised the plate coated with AF components (Intersmooth 7475Si SPC, International Paint Ltd., Gateshead, United Kingdom). As per the manufacturer's safety datasheet, the active ingredient of Intersmooth 7475Si SPC was 3.43% w/w copper pyrrhione and 37.45% w/w cuprous oxide. Both plates were stored at −70 °C before being submerged at JB-TMS. In the in situ experiment, the artificial plates were submerged 2 m below the water surface using 12-kg iron weights that were connected to the framework of the artificial plates and secured at the surface of the pontoons as well as on the seafloor with cinder blocks (Supplementary Fig. S1). The bacterial communities attached to the plate surfaces were collected seven times (at 1-week intervals, covering a total of 49 experimental days). The organisms attached to the submerged plates were collected in water to prevent damage via exposure to air. The organisms were stored in an ice cooler (approximately 4 °C) until arrival (5 min) to our laboratory at the Korea Institute of Ocean Science and Technology (Geoje, South Korea), where the sub-sample was prepared immediately. In addition, we collected 10-L samples of seawater (as ambient seawater) from a 2-m depth around the experimental site for analysing the bacterial communities and environmental factors.

2.2. Measurement of the physicochemical parameters

In the ambient seawater, temperature, salinity, pH and dissolved oxygen (DO) were monitored using the Multiparameter Water Quality Sonde (YSI 6920 Inc., Yellow Springs, OH, USA) on each sampling date at JB-TMS. A 100-mL aliquot of each sub-sample was filtered via a 47-mm glass fibre filter (GF/F, Whatman, Clifton, NJ, USA). Inorganic nutrient sources (dissolved inorganic nitrogen [DIN], dissolved inorganic phosphorus [DIP] and dissolved silica), dissolved organic carbon (DOC) and total bacterial counts were analysed as described in our previous report (Kang et al., 2021). The detailed method is described in Supplementary Information.

To perform a qualitative elemental analysis of the control and AF plates as per our previous analytical manual (Jung et al., 2015), we used energy-dispersive X-ray spectroscopy (EDS; X-Max model, Oxford, Abingdon, UK) that was attached to a field emission scanning electron

microscope (FE-SEM; JSM-7600F; Jeol, Tokyo, Japan). In brief, a sub-sample was obtained from each plate, and the desorbed paint particles in each plate were collected on a 0.2-μm isopore membrane filter (GTTP 04700, Millipore, Darmstadt, Germany); the filter was washed three times using 30 mL sterile distilled water. The harvested depth of the particles collected on the membrane was approximately 1 mm, and the samples were dried in a desiccator. After drying, the samples were coated with platinum and inspected using the FE-SEM instrument at 2000× magnification, with 8 mm working distance and a probe current exceeding 13 size. EDS analysis was performed by observing 10 areas for each sample, and the total number of EDS input counts used to analyse a sample was approximately 300,000 over 15 min. The accuracy of the results was ±1%.

2.3. 16S rDNA meta-barcoding analysis of bacterial communities

The bacterial communities were analysed in accordance with our established meta-barcoding analytical manual (Kang et al., 2021). Before collecting the bacterial communities, large-sized attached invertebrates and macroalgae were removed from both sides of the control and AF coatings. To analyse the biofouling bacterial communities attached to both plates, the plates were gently washed thrice with sterile filtered seawater. The bacteria were collected using a sterile rubber spatula by scraping both surfaces, and the collected samples were placed in sterile bottles. The samples were subjected to a pre-filtering step to remove the organic (i.e. spores, phytoplankton and zooplankton) and inorganic particles using a 3.0-μm polycarbonate membrane (TSTP 04700). Then, the bacteria in the samples were harvested on a 0.2-μm polycarbonate membrane (GTTP 04700) via filtration. Bacteria present in 500 mL ambient seawater were also harvested using a 0.2-μm membrane. Genomic DNA extraction and meta-barcoding analysis for bacterial communities were performed as described by Kang et al. (2021) and Supplementary Table S1, and the detailed method is described in Supplementary Information. The FASTQ data obtained from the MiSeq platform were deposited in the Sequence Read Archive database of National Centre for Biotechnology Information (NCBI) with the accession numbers SRR13132268–SRR13132289 (PRJNA 679337).

2.4. Statistical interpretation of the data

Alpha diversity (including Chao1, Shannon and Simpson diversity metrics) was plotted using a combination of customised R-values in RStudio (version 1.2.5042) using the vegan (Oksanen et al., 2019), ape (Paradis and Schliep, 2019) and ggplot2 packages (Wickham, 2016). A paired *t*-test was performed to examine differences in chemical composition between the control and AF coatings. *P*-values of <0.05 indicated significant differences. Pearson's correlation analysis was performed to examine the relationships among the measured parameters. All statistical analyses were performed using SPSS version 12 (SAS Institute Inc., Cary, NC, USA). To perform cluster analysis and non-metric multidimensional scaling (nMDS), the datasets of the common bacterial species as well as the environmental variables were assessed using several multivariate techniques. Prior to the analysis, the datasets were transformed to reduce the effect of skewness in the environmental dataset and to maximise the signal-to-noise ratio in the species dataset (Walker et al., 1997). The datasets of the environmental factors and the most abundant bacterial operational taxonomic units (OTUs) (each with a relative abundance of >1% in at least one sample) were primarily fourth-root transformed. Cluster analysis was accomplished by performing group average clustering using the Bray–Curtis similarity method on the most abundant bacterial OTUs. Using the ranked similarity matrix, an ordination plot was constructed via nMDS using PRIMER 6 (version 6.1.13). Hierarchical agglomerative clustering using the group average method was performed for the common bacterial OTUs (relative abundance >1%) on the basis of the groups selected from the nMDS analysis. To test the null hypothesis (no significant difference

among the groups as per the agglomerative clustering analysis), the similarities were examined using SIMPROF in PRIMER (Clarke, 1993).

To explore the functional annotation of prokaryotic taxa (FAPROTAX), we first selected the bacterial data using the results of the Venn diagram that appeared in each group, including the ambient seawater as well as control and AF coatings. We then performed FAPROTAX analysis using python collapse_table.py with the normalised OTU table (Louca et al., 2016). The FAPROTAX dataset (available at <http://www.zoology.ubc.ca/louca/FAPROTAX>) is a manually constructed database that is based on the cultured representatives of marine and freshwater microbiomes (Chun et al., 2019). The functional group abundances in each module were calculated by multiplying the calculated values (function tables) and the sum of OTUs belonging to each major module.

Correlation-based association networks have focused predominantly on the co-occurrence patterns and the potential keystone taxa in bacterial interconnectivity. Recently, network analysis was employed to elucidate the significant associations of host–parasite, host–bacteria and bacteria–environment factors (Berdjeb et al., 2018; Kang et al., 2021). To explore differences in the most abundant bacterial OTU networking between the control and AF coatings as well as ambient seawater, extended local similarity analysis (eLSA) (Xia et al., 2011) was used for seven datasets (obtained at 1-week intervals, covering 49 days of the total experimental period). The detailed method is described in Supplementary Information.

3. Results

3.1. Changes in the physicochemical characteristics of the seawater and chemical composition of AF coating

Variations in the physicochemical characteristics were observed in the ambient seawater. Water temperature and salinity gradually increased from 15.6 °C to 23.5 °C and from 32.5 °C to 33.3 °C, respectively (Fig. 1a). The DIN and DOC concentrations gradually decreased,

whereas the DIP concentration increased and showed an opposite pattern compared with the change in DIN concentration (Fig. 1b). The chemical characterisation of the surfaces of the control and AF coatings revealed the enrichment of certain elements, including C, O, Al and Si (with a relative mean weight of >5% for each element), which varied in the detached particles of both coatings (Fig. 1c). Some elements, such as Na, Mg, S, Cl, K, Ca, Ti, Cr, Fe, Cu and Zn, were found as minor elements (<5%), and the relative weights of Al, Cr, Cu and Zn in the AF coating were higher than those in the control plate. In particular, Cu was significantly different between the plates (*t*-test, -4.75 , $p < 0.05$).

3.2. Changes in the bacterial communities in AF coating

The total bacterial abundance remained the same in the ambient seawater, whereas it increased in the control and AF coatings. In particular, the bacterial abundance in AF coating increased rapidly (Fig. 1d). The meta-barcoding results are summarised in Table 1. The amplicons generated 28,371–190,000 read counts from the analysed samples. The average read counts in the ambient seawater and control and AF coatings were 132,861, 202,340 and 151,102, respectively. The OTUs and alpha diversity showed similar trends in terms of changes in read counts (Table 1). When the bacterial communities were classified using nMDS analysis, the communities in the ambient seawater were classified into a single group with 61% similarity (Fig. 2a). This group comprised the communities of *Alphaproteobacteria* (54%), *Flavobacteriia* (20%), *Gammaproteobacteria* (10%) and other groups. In the control plate, the bacterial communities were classified into two groups with 54% similarity (Fig. 2b). Group I was in the ‘initial settlement stage (start–14 days)’; in this group, *Alphaproteobacteria* were dominant, with a mean relative abundance of 98%. Group II was in the ‘irreversible adhesion stage (21–49 days)’. In this stage, the bacterial groups were more diverse than those in Group I; the diversities of *Gammaproteobacteria*, *Acidimicrobiia*, *Flavobacteriia* and *Deltaproteobacteria* were increased. In AF coating, the bacterial communities were classified into

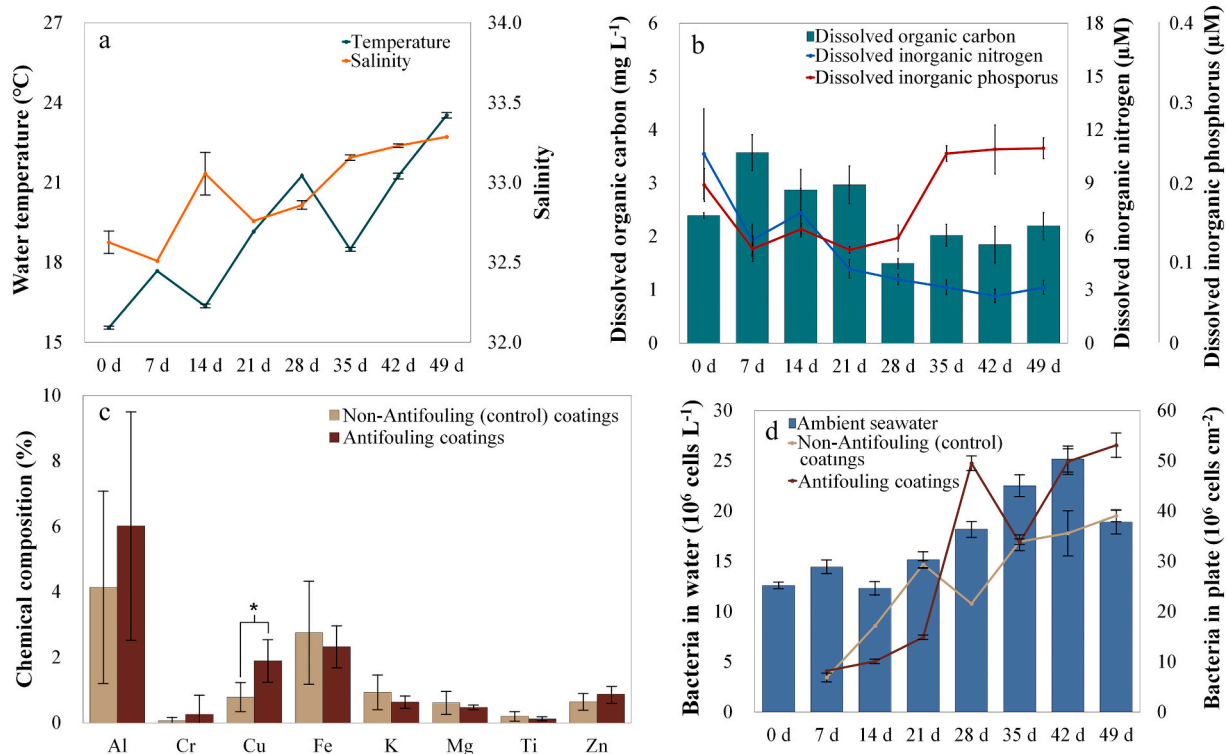


Fig. 1. Changes in the environmental factors, bacterial abundances and chemical composition. (a) Water temperature and salinity in the ambient seawater; (b) dissolved organic carbon, dissolved inorganic nitrogen and dissolved inorganic phosphorus in the ambient seawater; (c) chemical composition between the control and AF coatings and (d) total bacterial abundance among the ambient seawater and control and AF coatings. *, significant difference by *t*-test ($P < 0.05$).

Table 1

Summary of the read counts and operational taxonomic units (OTUs) of bacterial communities (16S rDNA V3–V4 regions) obtained via meta-barcoding analysis and the alpha-diversity indices of the experimental groups.

Samples	Group	During (day)	Total sequence (mean)	Read counts (mean)	OTUs (mean)	Chao1 (mean)	Shannon (mean)	Simpson (mean)
Ambient seawater	Group I	0–49	17,006,859–85,179,272 (59,283,604)	38,127–190,000 (132,861)	208–473 (355)	281.48–560.31 (434.78)	3.82–4.96 (4.32)	0.74–0.92 (0.84)
Non-Antifouling (control) coating	Group I	7–14	41,620,240–61,131,100 (51,375,670)	95,088–139,712 (117,400)	111–178 (144)	173.67–247.75 (210.71)	0.22–1.03 (0.63)	0.04–0.2 (0.12)
	Group II	21–49	66,837,557–100,911,546 (82,206,650)	151,399–224,791 (184,804)	462–840 (641)	534.71–937.11 (765.49)	4.25–6.41 (5.51)	0.84–0.95 (0.91)
Antifouling coating	Group I	7	165,080,310	357,594	114	141.05	0.29	0.06
	Group II	14–28	41,901,012–126,319,227 (85,288,258)	92,464–275,652 (186,195)	161–244 (199)	234.75–325.85 (267.87)	1.17–3.5 (2.26)	0.29–0.84 (0.55)
	Group III	35–49	12,879,985–39,595,296 (28,392,123)	28,371–87,863 (63,230)	169–241 (201)	190.67–266.74 (227.34)	4.27–4.43 (4.37)	0.87–0.9 (0.89)

The division of subgroups in each experimental group was obtained via non-metric multidimensional scaling (nMDS) analysis as presented in Fig. 2.

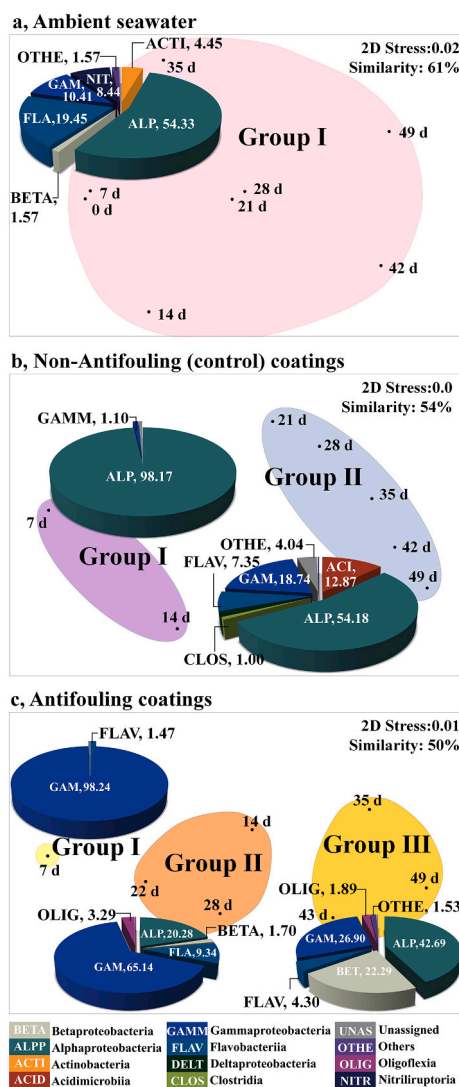


Fig. 2. Non-metric multidimensional scaling (nMDS) plot and Venn diagram in the ambient seawater (a), the control plate (b) and the anti-fouling coating (c). nMDS plot was analysed using the Bray–Curtis similarity method. The pie charts indicate the high-ranking taxonomic distribution at the class level of the bacterial community.

three groups with 54% similarity (Fig. 2c). Group I was in the stage of ‘biofouling formation by *Gammaproteobacteria* (start–7 days)’; in this group, *Gammaproteobacteria* were highly dominant, with a relative abundance of 98%. Group II was in the ‘early bacterial succession stage after biofilm formation (14–28 days)’. In this stage, the diversities of *Alphaproteobacteria*, *Flavobacteriia*, *Oligoflexia* and *Betaproteobacteria* were increased. Group III was in the stage of ‘formation of a stable bacterial community’, comprising *Alphaproteobacteria* (44%), *Gammaproteobacteria* (28%), *Betaproteobacteria* (19%), *Flavobacteriia* (5%) and *Oligoflexia* (3%). In particular, the predominant taxa were changed from *Gammaproteobacteria* to *Alphaproteobacteria* in this group.

The Venn diagram indicated that the bacterial species in the ambient seawater as well as control and AF coatings represented 622, 866 and 420 taxa, respectively (Fig. S2). In the control plate, bacterial OTUs overlapped in 170 taxa (20% of the total OTUs) and the number of OTUs in Group II (850) was higher than that in Group I (186). In AF coating, the number of OTUs in Groups I, II and III was 99, 222 and 358, respectively. The bacteria in the three groups overlapped in 77 taxa (18% of the total OTUs), and 14 taxa (3%) in Group I, 46 taxa (11%) in Group II and 178 (42%) taxa in Group III appeared as distinct taxa that did not overlap with the other groups.

3.3. Changes in the common bacterial taxa and the emergence of potential pathogenic bacteria

The common bacterial taxa in the analysed samples displayed a relative abundance of >1% in at least one sample. In the ambient seawater, the common bacterial taxa belonged to 31 OTUs and their cumulative relative abundance was 89% (Fig. 3a). The most dominant bacterium was OTU#002 (*Planktomarina* sp., *Alphaproteobacteria*, identity: 100%), with a mean abundance of 35%; the abundance of this taxon initially increased rapidly and then gradually decreased. OTU#006 (*Nitriliruptoria*, 85%) was the second most common taxon, with a mean of 8%; the abundance of this taxon was initially low and then continued to increase, thereby displaying a pattern opposite to that of OTU#002, OTU#022 (*Mesonina* sp., *Flavobacteriia*, 93%) and OTU#009 (*Yoonia* sp., *Alphaproteobacteria*, 98%) had mean abundances of 5% and did not exhibit any specific change in the pattern. In the control plate, in which the bacterial communities were classified into two groups, the common bacterial taxa belonged to 41 OTUs and their cumulative relative abundance was 84% (Fig. 3b). OTU#001 (*Pacificibacter* sp., *Alphaproteobacteria*, 99%) in Group I was the most common taxon, with a mean relative abundance of 94%. OTU#007 (*Phaeobacter* sp., *Alphaproteobacteria*, 98%) demonstrated a mean abundance of 1%, and the other 39 OTUs demonstrated mean abundances of <1%. In Group II, the

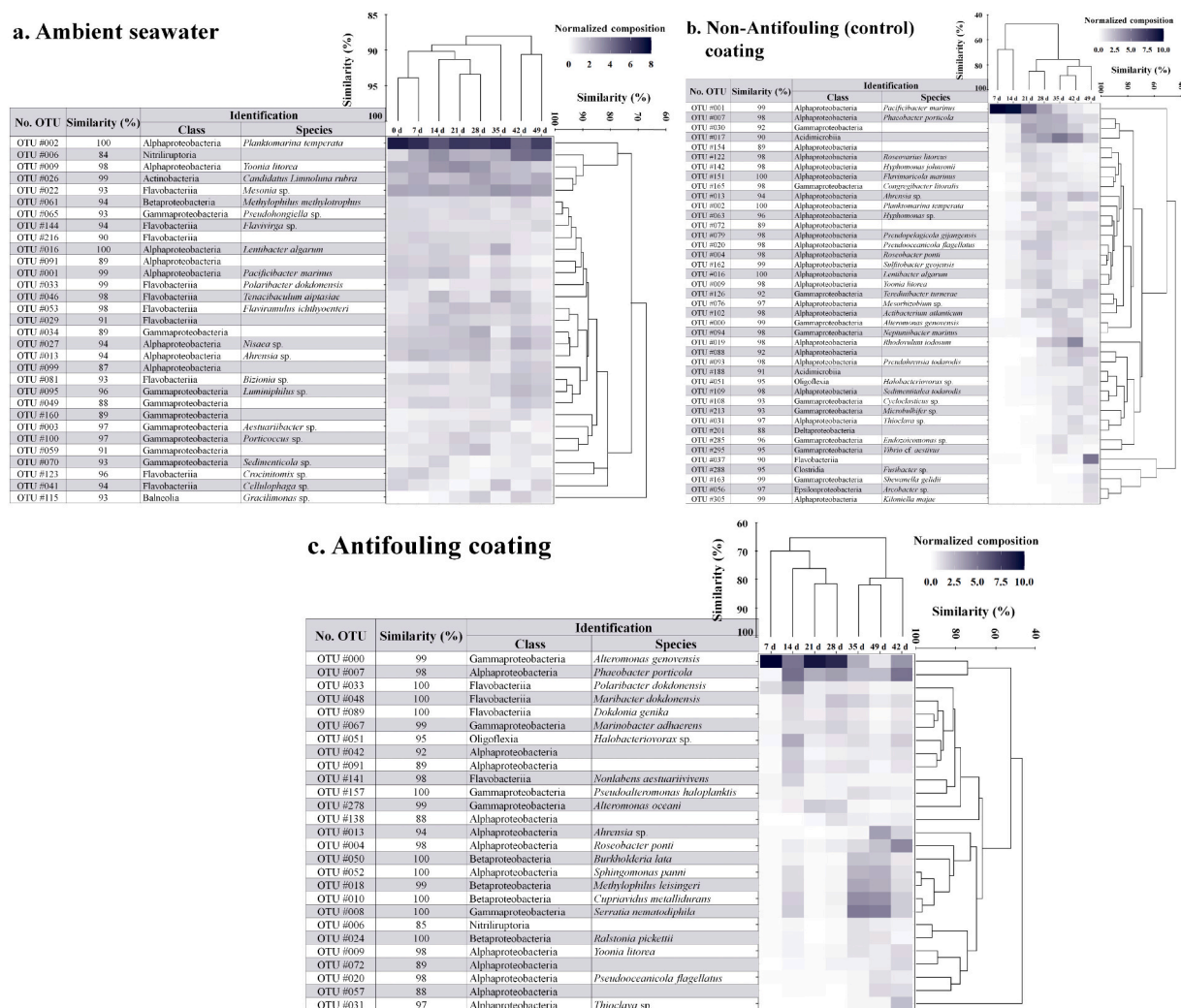


Fig. 3. Heatmap showing the most abundant bacterial operational taxonomic units (OTUs) Each OTU displayed a relative abundance of >1% in at least one sample in the ambient seawater (a), the control plate (b) and the anti-fouling coating (c). Heatmap displayed the fourth-root-normalised data. Hierarchical agglomerative clustering using the group average of the most abundant bacterial OTUs using the Bray–Curtis similarity method.

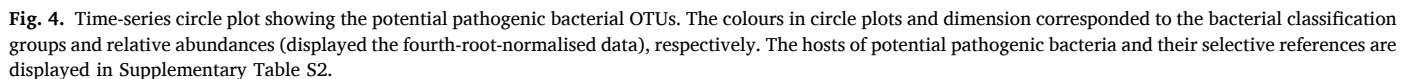
abundances of a bacterium in OTU#017 (*Acidimicrobia*, 90%), OTU#019 (*Rhodovulum* sp., *Alphaproteobacteria*, 98%) and OTU#013 (*Ahrensia* sp., *Alphaproteobacteria*, 94%) were observed to increase continuously, with mean abundances of 12%, 6% and 3%, respectively. The abundances of OTU#007 and OTU#030 (*Gammaproteobacteria*, 92%) initially increased and then decreased. OTU#037 (*Flavobacteriia*, 90%) had a relative abundance of <1% from the start to 42 days, but it increased rapidly to 25% on the last experimental day. The abundance of OTU#001, which constituted the predominant taxon in Group I, rapidly declined to 10%. In AF coating, in which the bacterial communities were divided into three groups, the common bacterial taxa belonged to 27 OTUs and the cumulative relative abundance of the common OTUs was 92% (Fig. 3c). In Group 1, which was independently classified on the 7th day, OTU#000 (*Alteromonas genovensis*, *Gammaproteobacteria*, 99%) was dominant, with a mean relative abundance of 97% and the other 26 OTUs demonstrated mean abundances of <1%. In Group II, the abundance of OTU#000 gradually decreased to 59%, whereas that of OTU#007 suddenly and rapidly increased to 16%. In Group III, the abundance of OTU#000 decreased to 8%, whereas that of OTU#008 (*Serratia* sp., *Gammaproteobacteria*, 100%), OTU#010 (*Cupriavidus* sp., *Betaproteobacteria*, 100%) and OTU#004 (*Roseobacter* sp., *Alphaproteobacteria*, 97%) suddenly and rapidly increased. The abundance of OTU#007 persisted at 16%.

A total of 41 OTUs belonging to the pathogenic bacteria were

identified, including *Actinobacteria* (2 OTUs), *Chlamydiae* (1 OTU), *Firmicutes* (1 OTU), *Proteobacteria* (36 OTUs) and *Tenericutes* (1 OTU) (Fig. 4, Supplementary Table S2). Among the *Proteobacteria*, *Gammaproteobacteria* accounted for the majority, with 24 OTUs belonging to *Alpha* (1), *Beta* (2) and *Epsilonproteobacteria* (1). In the ambient seawater and the control plate, 24 and 29 bacteria, respectively, were identified. In AF coating, 19 bacteria were identified and among them, OTU#050 (*Burkholderia* sp., 100%), OTU#010, OTU#024 (*Ralstonia* sp., 100%) and OTU#008 accounted for >1%. In particular, OTU#050, OTU#008 and OTU#010 were present continuously from the beginning of the experiment, increased rapidly to 16%, 4% and 23%, respectively, on the 35th day and remained high until the end of the experiment.

3.4. Co-occurrence of bacterial communities and bacterial putative function annotations

The network analyses of the ambient seawater and control and AF coatings revealed associated interactions with correlated common bacterial OTUs (Fig. S3 and Fig. 5). In the ambient seawater, there were associated with 22 nodes and 13 edges, which were simply interspersed between the nodes (Fig. S3a). In the control plate, there was a significant correlation, with 104 nodes and 534 edges (Fig. S3b). When divided into two sub-networks, the core-node of the first sub-network was PAMARINU, (OTU#001) (Fig. 5a and Supplementary Table S3) and was



As a result of FAPROTAX based on the common bacterial compositions (Fig. 6), the ambient seawater as well as control and AF coatings were defined in 17, 37 and 15 metabolic functions, respectively. The common metabolic functions were aerobic chemoheterotrophy and chemoheterotrophy, with a mean of 27% in the ambient seawater, 18% in the control plate and 32% in the AF coating. Each group had its specific metabolic characteristics: function of nitrogen fixation in the

Any natural or artificial substrate exposed to seawater is quickly fouled by bacteria (Dobretsov et al., 2013). Depending on the substrate, different microbial communities are found and play critical roles in the development of marine biofouling (Caruso, 2020; Sun et al., 2018). The biofouling bacterial community that accumulates on ship hulls is markedly affected by AF coatings (Winfield et al., 2018). In particular, early specific bacteria on ship hulls respond to chemical antagonism, induce the proliferation of specific biofilm bacteria and influence the interactions and recruitment of additional bacterial communities (Egan et al., 2001; Hadfield, 2011). In the present study, *Gammaproteobacteria* were initially dominant in AF coating, with a relative abundance of 98%, and were less taxonomically diverse in the early stages (Fig. 2). By contrast, *Alphaproteobacteria* were initially dominant in the control plate and demonstrated a relative abundance of 98%. Some *Gammaproteobacteria* are highly capable of growing on AF coating and possess substrate-specific chemical properties (Flach et al., 2017). After the initial selective attachment to the substrate, these bacteria secrete exopolysaccharides to temporarily adhere to the surface and form biofilms

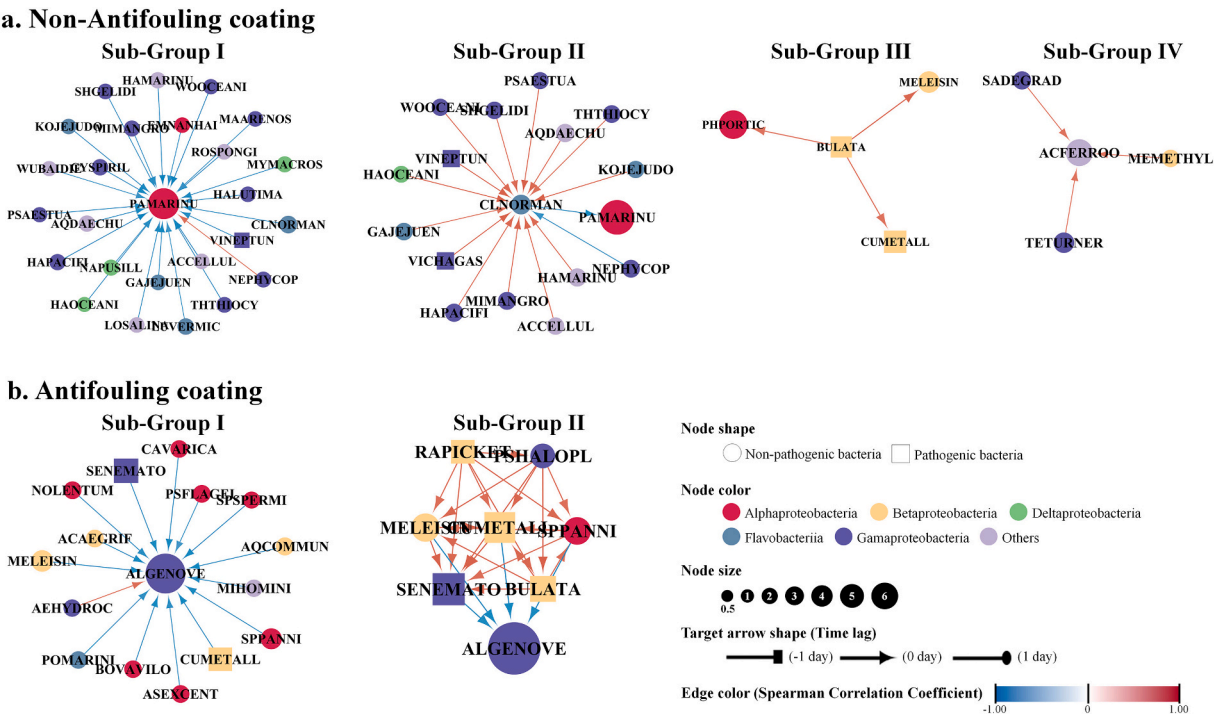


Fig. 5. Network analysis derived from the most abundant bacterial operational taxonomic units (OTUs). Zoomed images are the sub-networks associated with the control plate (a) and the anti-fouling coating (b). Total network figures are presented in Supplementary Fig. 3a, b and c. All nodes and edges were significantly correlated ($P < 0.01$, $Q < 0.05$). The identified bacterial species with OTU numbers are displayed in Supplementary Tables S3 and S4.

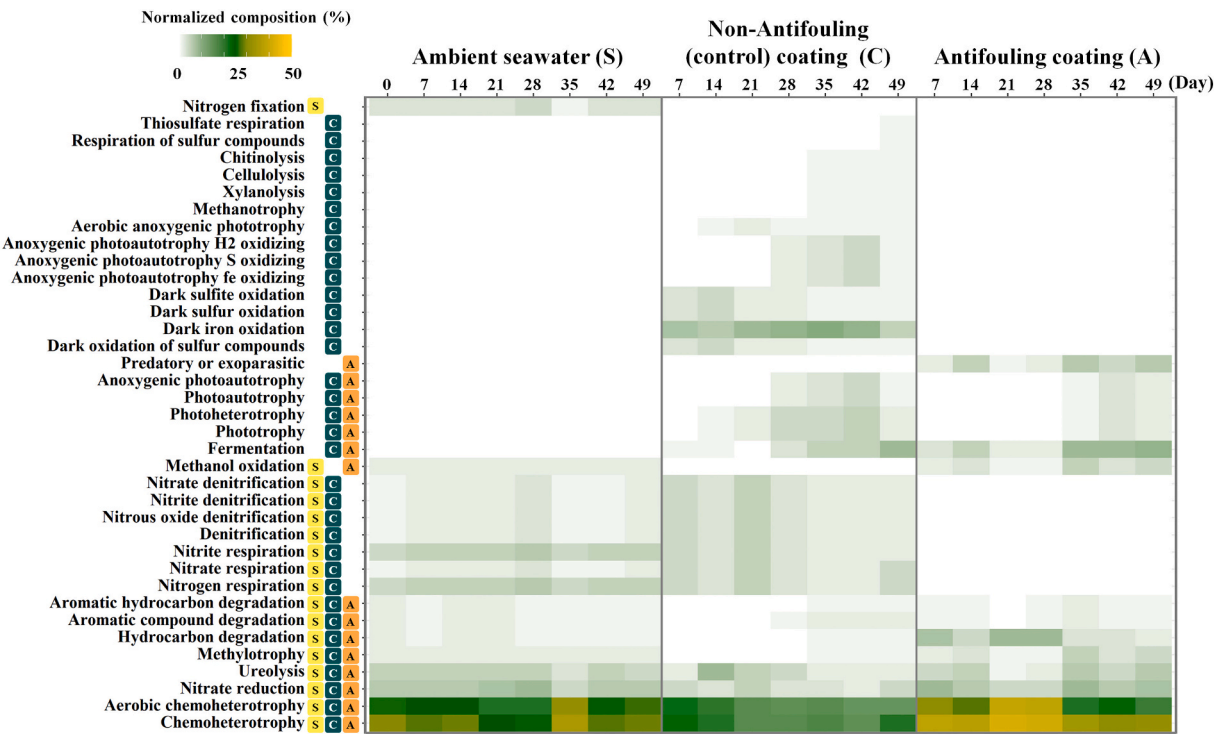


Fig. 6. Time-series heatmap showing the metabolic functions during the changes in bacterial community composition through the functional annotation analysis of the prokaryotic taxa (FAPROTAX). Heatmap displays the fourth-root-normalised data of the bacterial read counts.

(Krug, 2006; Muralidharan and Jayachandran, 2003). After the initial biofilm formation in AF coating, these fouling bacteria increase in diversity by neutralising the toxicity of the AF agent and result in the succession of other communities (Hadfield, 2011). In early-stage adhesion in the present study, *A. genovensis* in AF coating; this result is due to substrate specificity. *A. genovensis* was identified to be the early and primary coloniser in AF coatings with previous studies by Chen et al. (2013) and Vandecastelaere et al. (2008). The hull fouling bacterial biofilm is a highly organised community that includes a number of similar or homologous species (Costerton et al., 1995; Flemming et al.,

1996). *A. genovensis* is known to be tolerant to biocides (Chen et al., 2013; Vandecastelaere et al., 2008). Some *Alteromonas* members can produce biocidal compounds; the established bacterial biofilm community shields the late-arriving taxa from exposure to the biocides and hence, they are able to grow well on the biocidal surface (Egan et al., 2001; Hadfield, 2011).

In the mid and later stages, the abundance of *A. genovensis* decreased to 8% in AF coating, whereas the abundance of bacteria believed to be *Phaeobacter*, *Serratia* and *Cupriavidus* sp. rapidly increased. The increase in the abundance of *Phaeobacter* and *Serratia* sp. in the later stage may not only provide a selective habitat in AF coating but also inhibit the growth of other bacteria. Genus *Phaeobacter* and *Serratia* produce various secondary metabolites, particularly tropodithietic acid and bacteriocides (Berger et al., 2011; Matilla et al., 2016; Seyedsayamdost et al., 2011). Thus, the adhesion of the biofouling bacteria may be strongly influenced by specific antagonistic interactions among bacterial communities, which are likely to influence species' niche preferences (Burmølle et al., 2006). In the Fig. 5b, the rapid growth of *Cupriavidus* and *Serratia* sp. after biofilm formation by *A. genovensis* resulted in the growth of other bacteria that were presumed to be pathogenic. These are not only potential pathogens that attack human hosts but are also known to be resistant to metals or possess related metabolic activities, all of which are conducive for biofilm formation (Supplementary Table S2). In early stages in the control plate, the common taxon was *Pacificibacter* sp.; however, it changed to *Cloacibacterium* sp. over time. The interaction between these two taxa may be influenced by antagonistic relationships. *Cloacibacterium* sp. showed a positive correlation with potentially pathogenic *Vibrio* spp.. Among the pathogenic bacteria, the host of *V. chagasii* is scallops (i.e. *Patinopecten yessoensis*), whereas *V. neptunius* has various hosts, including bivalves, fish, shrimp and molluscs (Thompson et al., 2003). These potential pathogenic bacteria appear to have emerged because of the increased attachment and growth of invertebrates in the plate during the mid-stages of the experiment (refer to the pictures of control plate in Fig. S1).

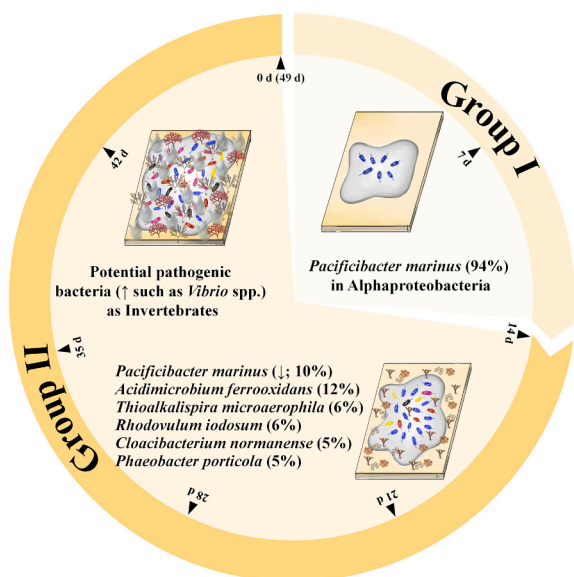
In the metabolic functional prediction using FAPROTAX, several sequences in the ambient seawater as well as control and AF coatings were assigned to chemoheterotrophy and aerobic chemoheterotrophy. These functions are categorised as broad ecosystem functions because they contain many activities performed by most microorganisms (Jung

et al., 2021; Rivett and Bell, 2018). In the control plate, most bacteria were found to have a high association with metabolic functions such as iron and sulphur oxidation. *Acidimicrobium* and *Rhodovulum* sp., which are associated with the metabolic functions of iron and sulphur oxidation and reduction, were increased from the 28th experimental day. In addition, their relative abundances are mainly high in acidic environments that are rich in iron and sulphur (Clark and Norris, 1996). Regarding the functions in AF coating, *Cupriavidus* sp. had exoparasitic metabolic functions. In addition, *Serratia* and *Methylophilus* sp. exhibited functions associated with methylotrophy. Thus, functional prediction established that most metabolic processes of the common taxa were found on specific harboured plates, which facilitated survival in iron- and copper-rich toxic environments. Furthermore, they provided a habitat for specific bacteria via their metabolism and biofilm-forming capacity. Therefore, the changes in the functional groups of specific bacterial populations were consistent with their substrate characteristics.

5. Conclusion

In the present study, the following evidences were obtained to support the accumulation of copper-tolerant bacteria and specific potential pathogenic bacteria in the biofilm coating (Fig. 7): (i) in early-stage adhesion, *A. genovensis* (Gammaproteobacteria) in AF coating and *Pacificibacter* sp. (Alphaproteobacteria) in the control plate were abundant. As mentioned in the Discussion, the intensive attachment of some Gammaproteobacteria (herein *A. genovensis*) on AF coating in the early stage may be associated with substrate-specific chemical responses on the coating. These specific bacteria associated with particular substrates exhibited specific stimulation and inhibition in response the released extracellular polymeric substances. This habitat may provide other bacteria an opportunity to promote or inhibit each other's growth. (ii) The abundance of *A. genovensis* decreased rapidly during the later stages in the AF coating, whereas that of *Phaeobacter*, *Serratia* and *Cupriavidus* sp. increased. This surge may provide a selective habitat in the biofilm formed by *A. genovensis* and may inhibit the growth of other bacteria. These selective organisms are not only potential human pathogens but are also known to be resistant to metals. On the control plate, the community was changed to *Cloacibacterium* sp., which had a strong

Non-Antifouling (control) coating



Antifouling coating

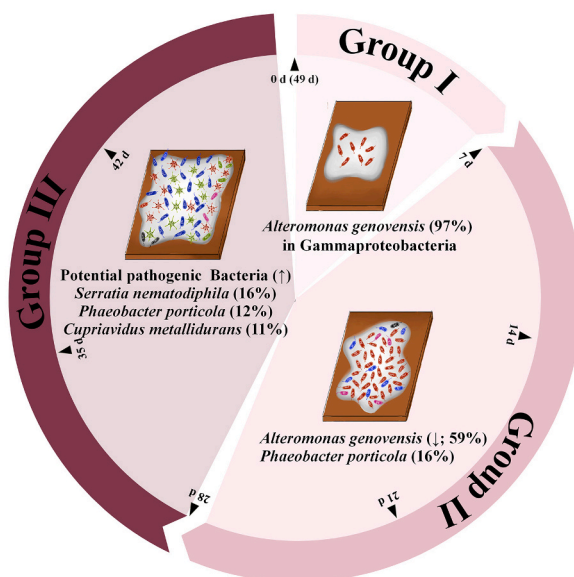


Fig. 7. Conceptual schematic diagram of the changes in common bacterial OTUs between non-antifouling and antifouling coatings.

connection with *Vibrio* spp. These *Vibrio* spp. appear to have emerged because of the increased attachment and growth of invertebrates, including scallops and bivalves. Therefore, in the in situ AF coating experiment, *A. genovensis* attachment was intensively increased on AF coating in the early stage and the bacterium responded to chemical antagonism, induced the proliferation of specific biofilm bacteria and influenced the interactions and recruitment of additional bacterial communities.

Ethical approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

CRediT authorship contribution statement

Kim, H.-J. and Park, J.S. contributed equally to this work. Jung S.W. and Kang J.H. designed the research plan; Kim H.-J. and Park J.S. performed the experiments and analysed the data; Park J.S., Jung S.W., Kang D., Shin K. and Lee T.-K. discussed the results; Kim H.-J., Park J.S. and Jung S.W. wrote the paper.

Declaration of competing interest

There are no conflicts of interests to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2021.112895>.

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